Weekly Progress Report

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Domain: Machine Learning / Industrial ML

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I. Overview

This week focused on baseline and comparative machine-learning models, validation, error analysis, and interpretation of model behaviour.

II. Achievements

1. Project 10: Trained baseline regression models and compared their performance using MAE, RMSE, and R^2.
2. Project 10: Evaluated whether process and time-derived features improved prediction quality and examinedfeature importance.
3. Project 6: Trained baseline RUL-regression models using engineered sensor features and sequence/windowrepresentations.
4. Project 6: Compared model errors across engines and different degradation patterns.
5. Applied time-aware validation principles to reduce leakage and obtain more realistic performance estimates.
6. Recorded model configurations and evaluation results for inclusion in the GitHub project report.

III. Challenges

1. Model performance varies significantly with forecast horizon and degradation stage.
2. Highly correlated industrial variables can make model interpretation difficult.
3. RUL prediction requires careful treatment of the target near failure and consistent evaluation across enginetrajectories.

IV. Learning Resources

1. scikit-learn regression and model-evaluation documentation.
2. References on predictive maintenance and Remaining Useful Life modelling.
3. Documentation for model-interpretability methods such as feature importance and SHAP.

V. Next Week's Goals

1. Tune the strongest baseline models.
2. Perform systematic feature selection and hyperparameter experiments.
3. Add explainability analysis and visualise important features.
4. Prepare final comparative results and limitations.

VI. Additional Comments

This week moved both projects from data preparation into measurable modelling work and established baseline performance for further improvement.